

Pseudo code of the PID Based Line Following Robot

START

INITIALIZE motor driver pins (ENA, ENB, IN1, IN2, IN3, IN4)

INITIALIZE IR sensors (S1, S2, S3, S4, S5)

SET baseSpeed = 60

SET PID constants

$K_p = 7$

$K_i = 0.001$

$K_d = 0.4$

SET integralLimit = 30

SET lastError = 0

SET integral = 0

SET samplingRate = 50 ms

SET motor direction to FORWARD

WAIT 5 seconds

WHILE robot is running

 GET current time

 IF current time – lastTime < samplingRate THEN

 CONTINUE loop

 END IF

 READ all IR sensors

 SET activeCount = 0

 SET weightedSum = 0

 IF S5 detects black line THEN

 weightedSum = weightedSum – 7

 activeCount = activeCount + 1

END IF

IF S4 detects black line THEN
 weightedSum = weightedSum - 3
 activeCount = activeCount + 1
END IF

IF S3 detects black line THEN
 weightedSum = weightedSum + 0
 activeCount = activeCount + 1
END IF

IF S2 detects black line THEN
 weightedSum = weightedSum + 3
 activeCount = activeCount + 1
END IF

IF S1 detects black line THEN
 weightedSum = weightedSum + 7
 activeCount = activeCount + 1
END IF

IF activeCount > 0 THEN
 error = weightedSum / activeCount
ELSE
 error = lastError
END IF

CALCULATE dt = elapsed time

integral = integral + (error × dt)

LIMIT integral within ± integralLimit

derivative = (error - lastError) / dt

correction = (Kp × error) + (Ki × integral) + (Kd × derivative)

maxSteeringValue = maxSpeed - baseSpeed

LIMIT correction within steering range

leftMotorSpeed = baseSpeed + correction

$\text{rightMotorSpeed} = \text{baseSpeed} - \text{correction}$

LIMIT motor speeds within minimum and maximum speed

SEND PWM signal to left motor

SEND PWM signal to right motor

MOVE robot forward

$\text{lastError} = \text{error}$

$\text{lastTime} = \text{currentTime}$

END WHILE

Flow Diagram

